REDUCING LLM GRADING INCONSISTENCY THROUGH QUALITY DIMENSION REFINEMENT: A MEASUREMENT THEORY APPROACH

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Abstract – *Automated grading helps manage instructor workload in engineering education, and large language models (LLMs) now make it feasible for short-answer responses. However, LLMs grade inconsistently: sampling variability flips identical responses between pass and fail in 8 to 20% of evaluations, undermining reliability. We address this with Quality Dimensions (QDs), binary rubric-derived criteria that are iteratively refined when grading is unstable. Applied to all 625 student responses across 25 questions in five embedded-systems laboratory assignments, QD refinement reduced the mean flip rate of unstable cases from 28.26% to 0.29% ($p < 0.001$, Cohen's $d = 2.00$) without destabilizing already-consistent responses. A small fraction (2.56%) resisted stabilization and was routed to human review; these disproportionately carried boundary instructor scores (68.8% versus 13.2%), suggesting refinement failure surfaces genuine ambiguity rather than system error. Consistency and correctness proved orthogonal: 12% of consistently graded responses did not match instructor judgment.*

Keywords: Automated short-answer grading; large language models; grading consistency and reliability; measurement theory; quantitative empirical evaluation

1. INTRODUCTION

Automated grading systems promise to reduce instructor workload while providing timely feedback to students. Large Language Models (LLMs) have emerged as tools for evaluating open-ended student responses, demonstrating capabilities in understanding context and assessing conceptual accuracy [1]. However, these systems face a fundamental reliability challenge: when the same student response is graded multiple times, the LLM may produce different pass-fail decisions. This inconsistency creates a measurement problem that undermines trust in automated assessment.

Consider an instructor who asks students to explain a technical concept. If an automated system grades the same response as "pass" on Monday but "fail" on Tuesday, neither the instructor nor the student can trust the assessment. This is not a hypothetical concern. Recent studies report that LLM grading decisions fluctuate in 10 to 20% of evaluations when temperature settings allow sampling variability [2, 3]. Such inconsistency is particularly problematic in engineering education, where assessments often involve technical explanations that require nuanced judgment about whether a student has demonstrated sufficient understanding.

The root cause lies in how LLMs generate responses. These models sample from probability distributions, meaning different runs can produce different outputs even with identical inputs. Setting the temperature parameter to zero forces deterministic behavior but may reduce the nuanced judgment needed for complex assessments [4]. The field needs methods that preserve judgment quality while reducing unwanted variability.

This paper presents a systematic approach to reducing LLM grading inconsistency through what we term *Quality Dimensions* (QDs). A Quality Dimension is a clear, binary criterion derived from a grading rubric. Rather than asking the LLM to produce a holistic grade directly, we decompose the assessment into explicit yes-or-no questions about the response. When inconsistencies arise, we treat them as signals that a criterion needs clarification, then iteratively refine the QD until grading stabilizes or the system reaches its iteration limit and routes the response to human review.

Our approach is grounded in measurement theory. Student understanding is a *latent variable* that cannot be directly observed [5]. We can only grade the visible text a student produces, which is an imperfect proxy for their actual comprehension. This means objective grading is likely impossible. However, *consistent* grading is achievable. Our goal is not to discover some "true" grade but to ensure that repeated assessments of the same response yield the same result.

We make three contributions. First, we introduce a formal framework that treats grading inconsistency as a refinable measurement problem rather than an inherent LLM limitation. Second, we provide empirical evidence from 625 engineering student responses showing that QD refinement resolves 81% of unstable cases ($d = 2.00$, $p < 0.001$), with the unresolved 2.56% correctly routed to human review. Third, we demonstrate that consistency and correctness are orthogonal properties, and that refinement failure correlates with instructor-uncertain boundary cases, suggesting that QD flagging serves as a quality filter beyond what baseline instability alone can provide.

2. BACKGROUND

2.1. The Latent Variable Problem in Educational Assessment

All educational assessment confronts a fundamental challenge: we want to measure student understanding, but understanding is a *latent variable* that cannot be directly observed [6]. We can only observe what students produce: written responses, code, or other artifacts. These observable outputs are imperfect proxies for the latent construct we wish to measure.

This creates two sources of measurement error. First, *encoding error* occurs when students understand a concept but cannot express it clearly in writing. Second, *decoding error* occurs when graders misinterpret what students meant. Both errors are unavoidable when grading written responses.

This theoretical framing has important implications for automated grading. If objective assessment is impossible even for human graders, we should not expect LLM-based systems to achieve objectivity either. Instead, the appropriate goal is *consistent subjectivity*: ensuring that the same response receives the same grade across repeated evaluations, even if that grade reflects a particular interpretive stance.

2.2. Consistency versus Correctness

Recent work has established that consistency and correctness are *orthogonal* properties in automated assessment [7]. A system can be consistent and correct, consistent but incorrect, or inconsistent. Reliability (consistency) does not guarantee validity (correctness): a system that always assigns "fail" is perfectly consistent but useless.

Our framework measures both dimensions. We quantify consistency through *flip rate*: the percentage of k repeated grading trials that disagree with the majority verdict. We quantify correctness through alignment with instructor judgments collected on a 1 to 10 Likert scale for a subset of 136 responses (Section 3.6). This dual measurement detects cases where a response is graded consistently but incorrectly.

2.3. Why Quality Dimensions?

Quality Dimensions draw from cognitive science research on conceptual spaces [8] and Formal Concept Analysis [9]. The key insight is that human similarity judgments are often non-metric: we assess whether features are present or absent rather than computing numerical distances. Traditional embedding-based approaches obscure grading rationale and shift the problem from "what makes a good answer" to "how do we convert text to numbers."

Quality Dimensions keep assessment in natural language. Each QD is a binary question: "Does this response demonstrate X?" The LLM answers yes or no for each dimension, and the final grade aggregates these judgments. This design is explainable (grades trace to specific criteria), refinable (unclear criteria can be rephrased without retraining), and aligned with how humans naturally assess quality through binary feature detection.

2.4. Related Work on LLM Grading Consistency

Several recent studies have examined LLM grading consistency. Zhang et al. [1] distinguished between absolute agreement (ICC2) and ranking consistency (ICC3), finding that ChatGPT achieved moderate absolute agreement (ICC = 0.50 to 0.73) but good ranking consistency (ICC = 0.75 to 0.89). Hackl et al. [4] tracked GPT-4 over 14 weeks, documenting short-term consistency (ICC = 0.999) but degradation over time (ICC = 0.944 by week 14).

Grévisse [2] found that rubric quality affected consistency more than temperature settings, suggesting that clearer criteria reduce variability. This finding motivates our approach: rather than constraining the LLM's sampling behavior, we clarify the criteria it evaluates.

Chu et al. [12] proposed GradeOpt, a multi-agent framework that iteratively refines grading guidelines through self-reflection. While conceptually related, GradeOpt optimizes for accuracy against human labels. Our approach differs in targeting consistency directly: we refine criteria to reduce flip rate rather than to match a particular set of human scores.

To the best of our knowledge, no prior work has combined iterative criterion refinement with formal measurement of both consistency and correctness as orthogonal dimensions. Our contribution fills this gap.

3. METHOD

3.1. Overview

Our approach consists of three components: (1) extracting Quality Dimensions from rubrics, (2) grading responses using QDs with repeated trials, and (3) iteratively refining QDs when inconsistencies arise. Figure 1 illustrates the pipeline, including the three possible outcomes: stable confirmation, successful improvement, or routing to human review.

3.2. Quality Dimension Extraction

Given a grading rubric R and a corpus of student responses C, we extract Quality Dimensions using an LLM. The prompt instructs the model to identify binary features that distinguish responses in the corpus. Each QD has:

- **Name**: Short identifier (e.g., "Technical Accuracy")
- **Definition**: Clear description of what presence means
- **Grading guidance**: Instructions for borderline cases

The extraction process examines both high-quality and low-quality responses to ensure QDs capture meaningful variation. Typically, 4 to 8 QDs are sufficient for short-answer questions. A representative extraction prompt is shown below; the full set of prompts (extraction, grading, and refinement) is omitted for space.

```
You are analyzing student responses
to the following question and rubric
to identify Quality Dimensions:
binary, natural-language criteria
that distinguish stronger from weaker
responses.
Question: {question}
Rubric: {rubric}
Sample responses: {high_quality},
{low_quality}
Return 4-8 Quality Dimensions. For
each, provide: (1) a short Name; (2) a
Definition stating what presence of the
feature looks like; (3) Grading guidance
for borderline cases. Each dimension
must be answerable as yes or no on
a single response without numerical
scoring.
```

3.3. Grading with Quality Dimensions

For each response r, we evaluate each QD independently. The LLM receives the question, the QD definition, and the response, then outputs a binary judgment (present/absent) with brief justification.

The final grade aggregates QD judgments. In our implementation, a response passes if it satisfies all "critical" QDs (those deemed essential by the rubric). This threshold can be adjusted based on assessment goals.

To measure consistency, we grade each response k times (we use $k = 10$). The *flip rate* is computed as:

$$\text{Flip Rate} = 100 - \frac{\max(\text{pass count}, \text{fail count})}{k} \times 100 \quad (1)$$

A flip rate of 0% indicates perfect consistency (all k trials agree). At $k = 10$, valid flip rates are 0%, 10%, 20%, 30%, 40%, and 50%. The *majority grade* is the more frequent outcome across k trials.

Algorithm 1 QD Refinement

Require: Responses C, QDs Q, baseline flip rate F_B, baseline majority grade G_B
Ensure: Refined QDs Q^* or FLAG_REVIEW

$Q^* \leftarrow Q$; $F^* \leftarrow \infty$
for $i = 1, 2$ **do**
    $F, G \leftarrow \text{GRADE}(C, Q, k{=}10)$
    if $F < F^*$ **then** $F^* \leftarrow F$; $Q^* \leftarrow Q$
    end if
    if $F^* = 0$ **then return** Q^*
    end if ▷ perfect
    if $F^* \leq F_B$ **and** $G = G_B$ **then return** Q^*
    end if
    $Q \leftarrow \text{REFINE}(Q, \text{FLIPPINGRESPS}(C, Q))$
end for
return FLAG_REVIEW

3.4. Iterative Refinement

When QD grading produces a higher flip rate than baseline or changes the majority grade, the system triggers refinement. The LLM analyzes which responses are flipping and which QDs contribute to the inconsistency, then applies one or more structural operators:

- **MERGE**: Combine dimensions that co-occur in $>$85% of cases. For example, "mentions energy" and "mentions charge" merged into "energy-per-charge relationship."
- **SPLIT**: Decompose an overly broad dimension when high internal variance is detected. For example, "explains pointers" split into "address concept," "operators," and "use cases."
- **ADD**: Introduce a dimension when unexplained grading variance suggests an uncaptured rubric aspect.
- **DROP**: Remove a non-discriminating dimension that appears in $>$95% or $<$5% of responses.
- **KEEP**: Retain a dimension unchanged when it already contributes to consistent grading.

Algorithm 1 formalizes the refinement logic. The algorithm tracks the best QD configuration and terminates early on perfect consistency or when flip rate no longer exceeds baseline. If two iterations cannot improve consistency, the response is flagged for human review.

3.5. Three-Dimensional Evaluation

We evaluate our approach along three orthogonal dimensions:

Dimension 1: Consistency. Measured by flip rate. Lower is better. A response is "stable" if its flip rate is 0%.

Dimension 2: Correctness. Measured by alignment with instructor judgments. An instructor rated a subset of

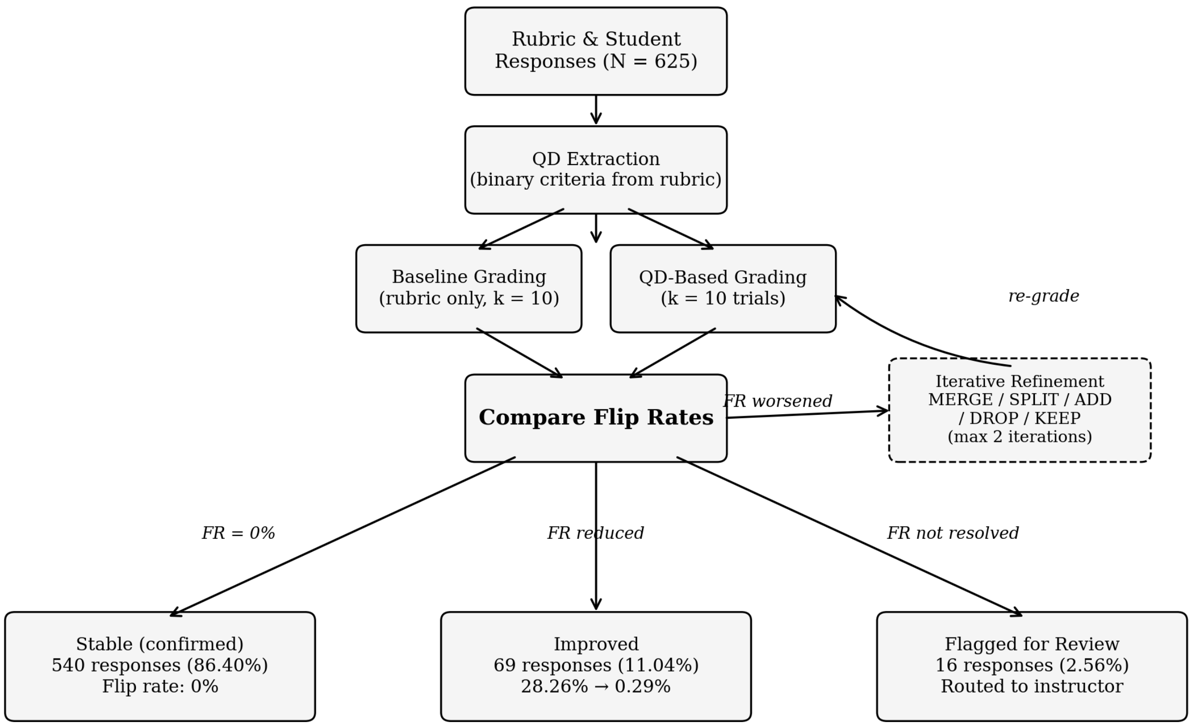

Figure 1: QD-based grading pipeline. Responses receive baseline grading ($k = 10$ trials), then QD grading ($k = 10$ trials). Consistency comparison routes responses to one of three outcomes. When QD grading worsens the flip rate, iterative refinement applies structural operators (MERGE, SPLIT, ADD, DROP, KEEP) and re-grades. If refinement does not improve consistency within two iterations, the response is flagged for human review.

136 responses on a 1 to 10 Likert scale, indicating the instructor’s confidence that the response would pass (Likert 6 to 10) or fail (Likert 1 to 3). Scores of 4 or 5 indicate uncertainty, in that the response was borderline and contained sufficient correct statements and incorrect statements or omissions that a grade of pass or fail was believed to be equally likely. This subset allows us to test whether consistency improvements come at the cost of correctness.

Dimension 3: Stability preservation. Measured by whether the method maintains already-stable responses. If baseline grading achieves 0% flip rate for a response, QD grading should not introduce instability for that response.

3.6. Dataset

We collected $N = 625$ student responses to 25 questions across five laboratory assignments from an undergraduate embedded systems course (MTE 241) at one Canadian institution. Each lab contained five conceptual explanation questions, with 25 responses sampled per question. Table 1 lists five representative questions from the first two labs.

Response lengths ranged from one to three paragraphs. An instructor independently rated a subset of 136 unique response texts on the 1 to 10 Likert scale, covering all 25 questions. Table 2 summarizes the dataset and Likert distribution.

3.7. Experimental Setup

All grading operations used gpt-oss-120b [13], an open-weight mixture-of-experts model with 116.8 billion parameters (5.1 billion active per token), deployed locally in GGUF format with native MXFP4 quantization ($\sim$65 GB). This model approaches the accuracy of proprietary reasoning models on standard benchmarks while permitting full reproducibility through local deployment [13]. Temperature was set to the model’s recommended default to allow sampling variation while maintaining coherent reasoning. Each response received $k = 10$ grading trials for both baseline and QD conditions.

Statistical analysis comprised:

- Paired t-tests comparing baseline and QD flip rates for

Table 1: Representative questions (Labs 1 and 2 of 5)

ID	Question
L1Q1	Explain what the HAL is and why it is used in a general embedded project.
L1Q2	Explain what the `io_putchar` function does and why it is needed.
L1Q3	Why did we have to use the debugger in this lab? Why didn't we just use `printf` to instrument our code?
L2Q1	Explain what happened when you changed PC to hold the function pointer to your print continuously function.
L2Q2	Explain what a system call is.

Table 2: Dataset summary

Characteristic	Value
Total responses	625
Number of labs	5
Questions per lab	5
Responses per question	25
Grading trials per response (k)	10
Instructor-rated subset	136
Likert 1 to 3 (Fail)	10 (7.35%)
Likert 4 to 5 (Uncertain)	18 (13.24%)
Likert 6 to 10 (Pass)	108 (79.41%)

unstable responses

- Cohen's d effect sizes with Kraft's [10] education-specific benchmarks ($d \geq 0.20$: medium; $d \geq 0.50$: large)
- McNemar's test for changes in instructor alignment
- 95% confidence intervals for all reported effect sizes

4. Results

We applied QD-based grading to all $N = 625$ responses. Every response, not only those showing baseline instability, received both baseline ($k = 10$ trials with rubric only) and QD-based grading ($k = 10$ trials with Quality Dimensions). This design allows direct comparison across the entire dataset and ensures that QD grading does not destabilize already-consistent responses.

4.1. Baseline Consistency

Baseline grading revealed that 540 of 625 responses (86.40%) were already consistent at 0% flip rate. The remaining 85 (13.60%) exhibited instability, with rates varying by lab. Lab 2 (Debugging & I/O) had the highest instability at 20.00%, while Lab 3 (RTOS Fundamentals) had the lowest at 8.00%. For the 85 unstable responses, the mean baseline flip rate was 26.82% (SD = 13.73%). Table 3 shows the per-lab breakdown.

Table 3: Per-lab baseline instability and QD outcomes

Lab	n	Unstable	Instab. Rate	Mean Flip Rate (%) After QD
1 (IDE & HAL)	125	15	12.00%	0.96
2 (Debugging)	125	25	20.00%	0.56
3 (RTOS)	125	10	8.00%	0.32
4 (Synchronization)	125	18	14.40%	0.80
5 (Integration)	125	17	13.60%	0.40
Overall	625	85	13.60%	0.61

Instability rate = proportion of responses with baseline flip rate $> 0\%$. Mean flip rate after QD is computed across all responses in the lab (stable + unstable). Per-lab rates range from 8.00% to 20.00%.

Table 4: Response outcomes after QD grading ($N = 625$)

Outcome	Count	%	Mean Flip Rate (%) Baseline	Mean Flip Rate (%) After QD
Stable (confirmed)	540	86.40	0.00	0.00
Improved	69	11.04	28.26	0.29
Flagged for review	16	2.56	11.00	18.75
All responses	625	100.00	3.65	0.61

Baseline = rubric-only grading ($k = 10$ trials). After QD = grading with Quality Dimensions ($k = 10$ trials). Flagged responses had lower baseline instability than improved ones but resisted resolution within two refinement iterations.

4.2. QD Refinement Impact

For the 540 baseline-stable responses, QD grading preserved their stability: none became unstable. In our dataset, stability preservation was complete. In practice, a small number of stable responses could become unstable under QD grading if the extracted dimensions introduce new ambiguity; such cases would also be routed to human review by the same degradation-detection mechanism described in Algorithm 1.

Of the 85 unstable responses, QD refinement resolved 69 (81.18%) and routed the remaining 16 to human review. Table 4 summarizes the outcomes with their associated flip rates.

For the 69 improved responses, the mean flip rate decreased from 28.26% (SD = 12.47%) to 0.29% (SD = 2.42%), a 98.97% relative reduction: $t(68) = 16.60$, $p < 0.001$, Cohen's $d = 2.00$, 95% CI $[1.59, 2.41]$. Using Kraft's [10] education-specific benchmarks, this represents a large practical effect. Figure 2 shows the per-response improvement.

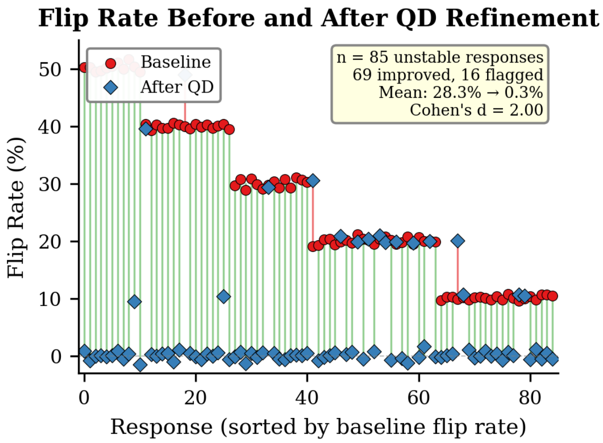

Figure 2: Paired flip rates for all 85 unstable responses, sorted by baseline flip rate. Red circles: baseline; blue diamonds: QD. Green lines connect improved responses ($n = 69$). Cohen's $d = 2.00$.

4.3. What Flagged Responses Reveal

The 16 flagged responses (2.56% of the dataset) are cases where QD refinement reached its two-iteration limit without reducing the flip rate below baseline. Rather than forcing a potentially incorrect automated resolution, the system stopped iterating and routed these responses to human review.

Three observations distinguish the flagged responses from the resolved ones.

First, **baseline instability did not predict flagging**. Flagged responses had a mean baseline flip rate of 11.00%, compared to 22.94% for the 69 successfully resolved responses. If we had used baseline flip rate alone to prioritize human review, these responses would have been ranked as *less* problematic, not more.

Second, **flagged responses disproportionately carried instructor-uncertain scores**. Of the 16 flagged responses, 11 overlapped with the instructor-graded sample. All 11 received Likert scores of 4 or 5 on the 10-point scale, placing them at the pass-fail boundary. That is, 68.8% of flagged responses carried boundary Likert scores (4 or 5), compared to 13.2% that scored in the same range across the full 136 instructor-graded responses. These are cases where even the human grader found the pass-fail decision difficult.

Third, **QD refinement failure correlates with correctness ambiguity, not criterion quality**. The responses that resisted automated resolution sat at a genuine boundary between pass and fail. No amount of criterion rephrasing can stabilize a response whose underlying quality is genuinely ambiguous. QD refinement correctly identified these cases and deferred them to human judgment.

The flagging mechanism thus provides a quality filter that goes beyond what baseline instability alone can offer: baseline instability identifies responses that fluctuate, while QD refinement failure identifies the subset whose fluctuation reflects genuine assessment ambiguity.

4.4. Consistency versus Correctness

Using the instructor-rated subset ($n = 136$), we examined whether consistency improvements came at the cost of correctness. Excluding uncertain cases (Likert 4 to 5, $n = 18$), we analyzed 118 responses. Of these, 108 achieved 0% flip rate after QD grading. Among these consistently graded responses:

- 95 (87.96%) were correctly aligned with instructor judgment
- 13 (12.04%) were consistent but incorrect

This confirms that reliability does not guarantee validity. A grading system can achieve perfect consistency while systematically assigning incorrect grades. Assessment systems must track both dimensions independently.

4.5. Instructor Alignment

Comparing LLM majority grades to instructor pass/fail classifications (excluding Likert 4 to 5, $n = 118$):

- Baseline agreement: 105/118 (88.98%)
- QD agreement: 102/118 (86.44%)

The discordant pairs were small ($b = 3$ where baseline was correct and QD was not, $c = 0$ where the reverse held). Because $b + c = 3$ falls below the threshold at which the chi-square approximation is reliable, we report McNemar's exact binomial test: $p = 0.250$, indicating no significant change in instructor alignment. QD refinement therefore maintained alignment while substantially improving consistency.

5. Discussion

5.1. Key Findings

Our results support four conclusions for our given dataset.

First, **QD grading is safe to apply to all responses**. For the 540 already-stable responses (86.40%), QD grading confirmed their consistency without introducing instability. The degradation-detection mechanism in Algorithm 1 provides a safety net: if QD grading were to destabilize a previously consistent response in other deployments, the same flagging logic would route it for human review.

Second, **QD refinement resolves the majority of grading inconsistency**. Of 85 unstable responses, 69 were resolved, reducing flip rates from 28.26% to 0.29% ($d = 2.00$, $p < 0.001$). The remaining 16 (2.56%) were routed to human review. The overall automated resolution rate of 97.44% suggests that most grading inconsistency stems from criterion ambiguity that natural language refinement can address.

Third, **refinement failure surfaces genuine correctness ambiguity**. Of the 16 responses routed to human review, 11 overlapped with the instructor-graded sample, and all 11 received Likert scores of 4 or 5. That is, 68.8% of flagged responses carried boundary Likert scores compared to 13.2% that scored in the same range across the full 136 instructor-graded responses. QD refinement failure detected what flip rate alone could not: responses where the pass-fail decision is genuinely uncertain even to human evaluators.

Fourth, **consistency and correctness are orthogonal**. Among consistently graded responses, 12.04% did not match instructor judgment. Assessment systems must measure and report both properties separately.

5.2. Implications for Practice

These findings have practical implications for engineering educators considering LLM-based grading.

Apply broadly, intervene selectively. QD grading, when applied to all responses, will likely not risk already-stable grades. The 2.56% flagged-for-review rate represents a minimal workload increase while providing a quality assurance mechanism.

Use refinement failure as a quality filter. When QD refinement cannot resolve inconsistency within two iterations, the response sits at a genuine assessment boundary. Educators can use this signal to prioritize human review on the responses that actually need it.

Do not rely on baseline instability alone. Flip rate alone does not predict which responses will resist automated resolution. Some responses with modest baseline instability (5 to 15%) proved harder to resolve than responses with high instability (30 to 40%).

5.3. Model-Agnosticism of the QD Framework

A natural question is whether the results obtained with gpt-oss-120b transfer to other LLMs. The QD framework treats the grading model as a black-box judge: it depends only on the model being able to answer binary natural-language questions about a response. The refinement logic itself (MERGE, SPLIT, ADD, DROP, KEEP) is independent of any specific model and operates on the criterion text rather than on model internals. We therefore expect the QD logic to be largely model-agnostic. The empirical quantities, however, including the 81% resolution rate and the 2.56% flag rate, are model-specific and may shift with stronger or weaker base models. Replication across models is left as future work.

5.4. Ethical Considerations

The use of LLMs in student assessment raises questions that extend beyond technical performance. We do not argue that LLMs should replace human grading. Rather, given that automated grading is increasingly deployed in practice, consistency is a measurable property that should be addressed before such systems affect student outcomes.

Three features of our approach speak to these concerns. First, the flagging mechanism preserves human oversight by design: the system defers ambiguous cases rather than forcing automated decisions at the pass-fail boundary. Second, QD-based grading is explainable: each grade traces to specific binary criteria, supporting transparency to students and a basis for appeal. Third, the instructor remains the responsible grader; the LLM functions as a triage tool, not a decision-maker.

We acknowledge limitations to this framing. We did not test whether LLM judgments vary systematically across writing styles, including those of non-native English speakers, which is an important fairness question for future work. Our study used anonymized response data with no identifying information generated through analysis or dissemination, and qualified for research ethics board exemption under the Tri-Council Policy Statement (TCPS 2); the exemption basis is stated in the Acknowledgements.

5.5. Limitations

Several limitations constrain interpretation of these results.

Single institution. All data came from one course at one institution. Generalization to other contexts, disciplines, and student populations requires replication.

Limited question types. We tested only conceptual explanation questions. Technical problem-solving, design, or coding questions may behave differently under QD-based grading.

Instructor subjectivity. Our correctness ground truth is one instructor's judgment on a subset of 136 responses. Inter-rater reliability for the Likert scores was not assessed.

Model specificity. Results were obtained with gpt-oss-120b [13] and may not transfer to other LLMs or future model versions, as model behavior evolves over time [4]. As discussed above, the QD logic is expected to be model-agnostic, but resolution and flag rates are empirical quantities tied to this specific model.

5.6. Future Work

Several directions merit investigation: testing across multiple institutions and disciplines, replicating the resolution and flag rates across LLMs of different scales to test the model-agnosticism claim, comparing QD-based grading to fine-tuned models and multi-agent approaches such as GradeOpt [12], increasing k to resolve borderline cases at the minimum detectable instability threshold, and characterizing which response properties predict QD refinement failure.

6. Conclusion

This paper introduced Quality Dimensions as a method for reducing LLM grading inconsistency in engineering education. Applied to 625 responses, QD grading confirmed the stability of 540 already-consistent responses while resolving 69 of 85 unstable cases, reducing their mean flip rate from 28.26% to 0.29% ($d = 2.00$, $p < 0.001$). The remaining 16 (2.56%) were routed to human review.

Our central finding is that grading inconsistency signals unclear criteria, not inherent LLM limitations. When refinement cannot stabilize a response, this failure is itself diagnostic: of the 16 flagged cases, 11 had instructor Likert scores and all fell at the 4 to 5 boundary, a rate of 68.8% compared to 13.2% in the overall graded set. Future work should investigate whether QD refinement failure predicts assessment boundary cases across institutions and disciplines.

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This study relied exclusively on the secondary use of anonymized student response data with no identifying information generated through analysis or dissemination. It therefore qualified for exemption from research ethics board review under the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2, 2022), Chapter 2, Article 2.4 (`https://ethics.gc.ca/eng/tcps2-eptc2_2022_chapter2-chapitre2.html`).

Disclosure Statement

The authors used a large language model (Claude, Anthropic) for language editing and formatting assistance during the preparation of this manuscript. All research design, data collection, analysis, interpretation, and conclusions are the authors' own, and no generative AI tool was used to generate or analyze the study's results. Note that gpt-oss-120b is employed as the object of study (the grading model under evaluation), not as a writing aid, as described in Section 3.6 and the Experimental Setup.

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